

C# BACKEND TRACK | OOP PHASE

Task 1

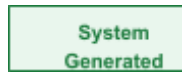
System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:



System Generated

The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as `List.Count + 1`.



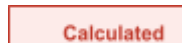
User Input

The user types this value directly at the keyboard when prompted by the system.



Default Value

A fixed value set by the code when the object is created. The user never chooses it. Example: `isBooked = false`, `status = "Active"`.



Calculated

Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.



From List

The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.



Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: **Book**

Property Name	Data Type	Source Tag
BookID	int	System Generated
BookTitle	string	User Input
BookAuthor	string	User Input
BookTotalCopies	string	User Input
BookAvailableCopies	int	Default Value

Entity Name: **Member**

Property Name	Data Type	Source Tag
MemberID	int	System Generated
MemberName	string	User Input
MemberEmail	string	User Input
MemberPhone	string	User Input
MemberRegistareDate	string	System Generated
IsMemberActive	bool	Default Value

Entity Name: **Loan**

Property Name	Data Type	Source Tag
LoanID	Int	System Generated
BookId	int	From List
MemberID	int	From List
BorrowDate	string	System Generated
BookCopyReturnDate	string	System Calculated
ReturnDate	string	System Generated
FineAmount	Double	System Calculated
Status	string	Default Value

Entity Name: **Reservation**

Property Name	Data Type	Source Tag
reservationId		System Generated
BookID		From List
MemberID		From List
reservationDate		System Generated
Status		Default Value

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest		
Property Name	Data Type	Source Tag (fill in)
guestId	int	System Generated
guestName	string	User Input
guestPhone	string	User Input
guestEmail	string	User Input
guestNationality	string	User Input
registrationDate	string	System Generated

Room		
Property Name	Data Type	Source Tag (fill in)
roomId	int	System Generated
roomNumber	string	User Input
roomType	string	User Input
floorNumber	int	System Generated
nightlyPrice	decimal	System Calculated
isAvailable	bool	Default Value

Booking		
Property Name	Data Type	Source Tag (fill in)
bookingId	int	System Generated
guestId	int	From List
roomId	int	From List
checkInDate	string	User Input
checkOutDate	string	User Input
numberOfNights	int	System Calculated
totalPrice	decimal	System Calculated
status	string	Default Value

CheckInRecord

Property Name	Data Type	Source Tag (fill in)
<code>recordId</code>	int	System Generated
<code>bookingId</code>	int	From List
<code>guestId</code>	int	From List
<code>roomId</code>	int	From List
<code>actualCheckIn</code>	string	System Generated
<code>actualCheckOut</code>	string	System Generated
<code>finalAmount</code>	decimal	System Calculated